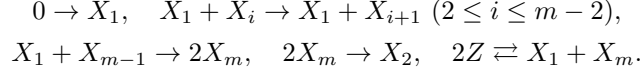


Exact Diffusion Design for Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

Topology-wide all-spectrum localization and exponent-optimal stationary heterogeneity trade-offs

Concise theorem summary — external audit draft

Family/realization space. For every $m \geq 3$, the binary-complex mass-action topology $\hat{N}_m$ has species $X_1, \dots, X_m, Z$ and reactions



The chain range is empty for $m = 3$. With $n = m + 1$, there are n species and $n + 1$ reactions. The semipositive conservation vector is $c = (0, 4, \dots, 4, 2, 1)^T$, $\text{rank } \Gamma_m = m$, and

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

The exact two-dimensional kernel and the $m + 2$ columns give the stated rank by rank-nullity; the maximal minor on rows $X_1, X_3, \dots, X_m, Z$ and columns $R_2, \dots, R_{m-2}, R_a, R_b, R_+$ has determinant $4(-1)^m$. Every positive steady flux is $(a\mathbf{1}_m, b, b)$, and every positive-equilibrium Jacobian is exactly $J_m(a, b, H) = A_m(a, b)H$ with $a, b > 0$ and $H \succ 0$ diagonal. Conversely every such triple is physically realized.

Topology-wide localization. For every positive realization and every principal species set I with $0 < |I| < m$, $J_{I,I}$ is Hurwitz. Every strongly connected diagonal block is either a negative singleton, one of the two stable long cycles, or a non-singleton principal block of the boundary triad $\{X_1, X_m, Z\}$. The triad satisfies the cubic Routh–Hurwitz inequalities coefficientwise. At $b = 2a$, the deleted edge $X_1 \rightarrow X_m$ lies on neither long cycle and can only refine the boundary decomposition. For $m = 3$, the two cycles $\{X_1, X_2\}$ and $\{X_2, X_3\}$ and all remaining principal subgraphs are checked directly; the chain classification is used for $m \geq 4$. In contrast,

$$(-1)^m \det J_{\{X_1, \dots, X_m\}, \{X_1, \dots, X_m\}} = -2a^{m-1}b \prod_{i=1}^m h_i < 0.$$

Hence, throughout every positive realization,

$$\min \{|I| : \emptyset \neq I \subseteq [n], \alpha(J_{I,I}) > 0\} = m = n - 1.$$

Every smaller subsystem is stable against both real and complex instability.

Principal-minor diffusion-ray theorem. Let $n \geq 2$ and $a_I = (-1)^{|I|} \det J_{I,I}$. Assume $\det J = 0$, $a_I > 0$ for $|I| \leq n - 2$, and $\sum_{|I|=n-1} a_I > 0$ (in particular these coefficient hypotheses follow from a simple zero and otherwise stable spectrum together with stable lower principal blocks). Then for $D = \text{diag}(d_1, \dots, d_n)$ with $d_j > 0$

$$\det(sD - J) = s(\beta_1 + \beta_2 s + \dots + \beta_n s^{n-1}), \quad \beta_k > 0 \quad (k \geq 2).$$

A nonzero stationary threshold exists iff $\beta_1 < 0$; it is unique and simple. Moreover $J - sD$ has a positive real eigenvalue exactly for $0 < s < s_*$. This does not rule out nonreal wave instability outside that band. The network application is a one-bad-minor corollary, not a hypothesis of the general theorem. Here scalar simplicity is first proved for the dispersion root; ordinary algebraic simplicity of zero for $J - s_* D$ follows separately from $\chi'_{s_*}(0) > 0$.

Exact network diffusion law and sharp linear contrast. For $D = \text{diag}(d_1, \dots, d_m, d_Z) \succ 0$, the complete order- $(n - 1)$ omission table has one negative term (omit Z), positive weight 16 when an interior X_j is omitted, and zero when X_1 or X_m is omitted. Thus

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j}$$

is necessary and sufficient for a stationary threshold on the ray sD . For every H in the explicitly defined homogeneously stable domain, the sharp nonattained contrast infimum is

$$T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1};$$

at $H = I$ it is $8(m - 2)$. Every stationary crossing of this topology obeys $\chi_D \chi_H > 8(m - 2)$. Here χ_H measures equilibrium concentration-scale separation; it is not interpreted as a universal measure of biological expense or implausibility.

Here $c^T J = 0$ and homogeneous stability on $c^\perp$ imply $\text{rank } n - 1$, an algebraically simple conservation zero, and a positive order- $(n - 1)$ characteristic coefficient. Thus the general diffusion-ray hypotheses hold. The criterion is independent of a, b , but the threshold is $s_*(a, b, H, D)$ and generally depends on them.

Exact unit-equilibrium stable design. At $a = b = 1$, $H = I$, write $K_i = 91m - 181 - i$ and take

$$d_1 = \frac{23}{63}, \quad d_i = K_i^{-1} \quad (2 \leq i \leq m-1), \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}.$$

Write the full critical vectors as $r = (r_1, \dots, r_m, r_Z)^T$ and $\ell = (\ell_1, \dots, \ell_m, \ell_Z)^T$. The exact contrast is $23(91m - 183)/63$. A 35-term homogeneous half-plane certificate and a 77-term spatial certificate exclude every nonzero closed-right-half-plane first-mode root. Algebraic simplicity at the equality point follows from

$$\frac{d}{d\lambda} \det(\lambda I - A_m + D_m) \Big|_{\lambda=0} = -\frac{163}{45} \ell^T r > 0.$$

Thus the first-mode kernel and cokernel are one-dimensional; exact critical vectors also give transversality $\eta_m > 0$. The homogeneous base case $m = 3$ is checked from $q_3 = (\lambda + 7)(\lambda^2 + 5\lambda + 2)$; the 35-term root exclusion applies for $m \geq 4$.

On the fixed integrated-mass class, the quadratic field satisfies

$$A_m w_0 = -\frac{1}{4} B(r, r), \quad c^T w_0 = 0, \quad (A_m - 4D_m) w_2 = -\frac{1}{4} B(r, r).$$

The cubic coefficient is

$$c_m = \frac{\ell^T \{B(r, w_0) + \frac{1}{2} B(r, w_2)\}}{\ell^T r}.$$

This is the one-dimensional center-manifold normal-form coefficient in the adjoint-projection coordinate; setting the reduced vector field to zero gives the stationary Lyapunov–Schmidt equation. Spatial reflection $\xi \mapsto \pi - \xi$ sends $r \cos \xi$ to its negative and acts as $A \mapsto -A$ on an equivariant center manifold. Hence the reduced field is odd and the two nonzero branches are exchanged by reflection. Because their perturbations tend to zero in $H_N^2 \hookrightarrow C^0$ from $\mathbf{1}$, both branches are componentwise strictly positive for sufficiently small $\mu > 0$ at each fixed m . For the stationary onset linearization $\mathcal{L}_0 = A_m + D_m \partial_{\xi\xi}$, fixed-mass invariance follows from $c^T A_m = 0$ and the Neumann boundary conditions. The cosine blocks are $A_m - k^2 D_m$: the $k = 0$ block is invertible on $c^\perp$, $k = 1$ has the certified simple zero, and $k \geq 2$ is invertible. The factorization

$$A_m - k^2 D_m = -k^2 D_m (I - k^{-2} D_m^{-1} A_m), \quad \|(A_m - k^2 D_m)^{-1}\| = O(k^{-2}),$$

maps compatible L^2 data into H_N^2 and proves that the stationary fixed-mass linearization is Fredholm of index zero, with kernel $\text{span}\{r \cos \xi\}$, cokernel $\text{span}\{\ell \cos \xi\}$, and nonzero transversality pairing $(\pi/2) \ell^T D_m r$. A four-factor recurrence and a shifted-polynomial certificate prove $c_m < 0$ for every $m \geq 3$. The two supercritical positive branches are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.

Stable diffusion–equilibrium trade-off. Let $\nu = m - 2$,

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \quad L_0 = \kappa_\nu / \sqrt{\nu}, \quad L_1 = 90\nu / (90\nu + 1),$$

and for $L \in [L_0, L_1]$ set

$$h_1 = h_m = h_Z = 1, \quad h_i = \frac{K_i}{LK_{i-1}}, \quad H_m(L) = \text{diag}(h), \quad D_m^{\text{phys}}(L) = H_m(L) \Delta_m,$$

where Δ_m is the effective unit profile above. The physical equilibrium is $H_m(L)^{-1} \mathbf{1}$. Under $\hat{x} = H_m(L)x$, the scaled PDE retains the parameter as $\partial_t \hat{x} = H_m(L) \{f_m(\hat{x}) + (1 - \mu) \Delta_m \partial_{\xi\xi} \hat{x}\}$; at $\mu = 0$ its stationary bracket and critical right vector are unchanged. The interval is nonempty in every dimension. The transformed left vector $\tilde{\ell}(L) = H_m(L)^{-1} \ell$ satisfies

$$\tilde{\ell}(L)^T H_m(L) \Delta_m r = \ell^T \Delta_m r < 0, \quad \tilde{\ell}(L)^T r < 0,$$

so the crossing coefficient is positive. The kernel of the row-scaled first-mode operator remains $\text{span}\{r\}$; the nonzero pairing $\tilde{\ell}(L)^T r$ excludes a generalized eigenvector, so the scaled zero is algebraically simple. A 22-term homogeneous certificate for $\nu \geq 2$, a direct Routh–Hurwitz calculation for $\nu = 1$, an 84-term spatial certificate, and a gauge-corrected cubic comparison prove a primary supercritical branch that is locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space for every certified L . The normalized branch is positive for small $\mu > 0$ by $H_N^2 \hookrightarrow C^0$, and the positive diagonal physical rescaling preserves componentwise positivity. The physical contrasts are

$$\chi_D(L) = \frac{23}{63} 91\nu L, \quad \chi_H(L) = \frac{91\nu - 1}{91\nu L}, \quad \chi_D \chi_H = \frac{23(91\nu - 1)}{63}.$$

The product is independent of L and equals the unit-equilibrium stable-profile contrast,

$$\chi_D(L) \chi_H(L) = \chi_D^{\text{unit}}(m) = \frac{23(91m - 183)}{63}.$$

Thus the family reallocates a fixed contrast product without reducing it. Throughout the certified interval, $\chi_D(L) > \chi_H(L)$, so $\max(\chi_D, \chi_H) = \chi_D(L)$ is uniquely minimized within this family at $L = L_0$. At $L = L_0$,

$$\chi_D = \frac{2093}{63} \kappa_\nu \sqrt{\nu}, \quad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu} \frac{91\nu - 1}{91\nu},$$

so both, and hence their maximum, are $\Theta(\sqrt{m})$, compared with $\Theta(m)$ for the unit-equilibrium stable profile. The lower endpoint is the boundary of the present sufficient homogeneous modulus estimate for $\nu \geq 2$; it is not a proved intrinsic dynamical boundary. Since every stationary crossing of $\tilde{N}_m$ satisfies $\max(\chi_D, \chi_H) > \sqrt{8(m-2)}$, exponent 1/2 is optimal among stationary-crossing realizations of this topology. No constant-optimal or complete Pareto-frontier claim is made.

PDE conclusion. On fixed-mass L^2 , positive diagonal Neumann diffusion with domain H_N^2 is sectorial with compact resolvent, and the reaction Jacobian is a bounded multiplication perturbation. The high-mode spectral bound tends to $-\infty$, so only finitely many low-mode gaps require analytic continuation along the branch. The half-order phase space is H^1 , and the quadratic mass-action Nemytskii map is smooth into L^2 . The conservation restriction removes the homogeneous zero and the Neumann interval has no continuous translation mode. Strict branch-spectrum negativity therefore yields local exponential convergence for sufficiently close nonnegative initial data in the same mass class.

Most failure-prone points. External checking should focus on: (i) exhaustion of principal SCCs, including $b = 2a$; (ii) strict equality cases in the 22/35/77/84-term certificates and the direct $\nu = 1$ cubic; (iii) the principal-minor derivative monotonicity; (iv) the equilibrium-scaled fixed-mass gauge; and (v) the cubic comparison $N_m(L) > 1/200$.

Scope. The family is synthetic, rank deficient with one semipositive conservation law, and all-mobile. Results are local for each fixed dimension. No arbitrary-data global boundedness, global attraction, constant-optimal stable frontier, biochemical realization, cross-diffusion, or immobile-species theorem is claimed. Here Turing bifurcation means a stationary diffusion-driven crossing of a nonconstant Neumann mode on a fixed interval; intrinsic finite-wavelength selection under domain enlargement is not claimed.